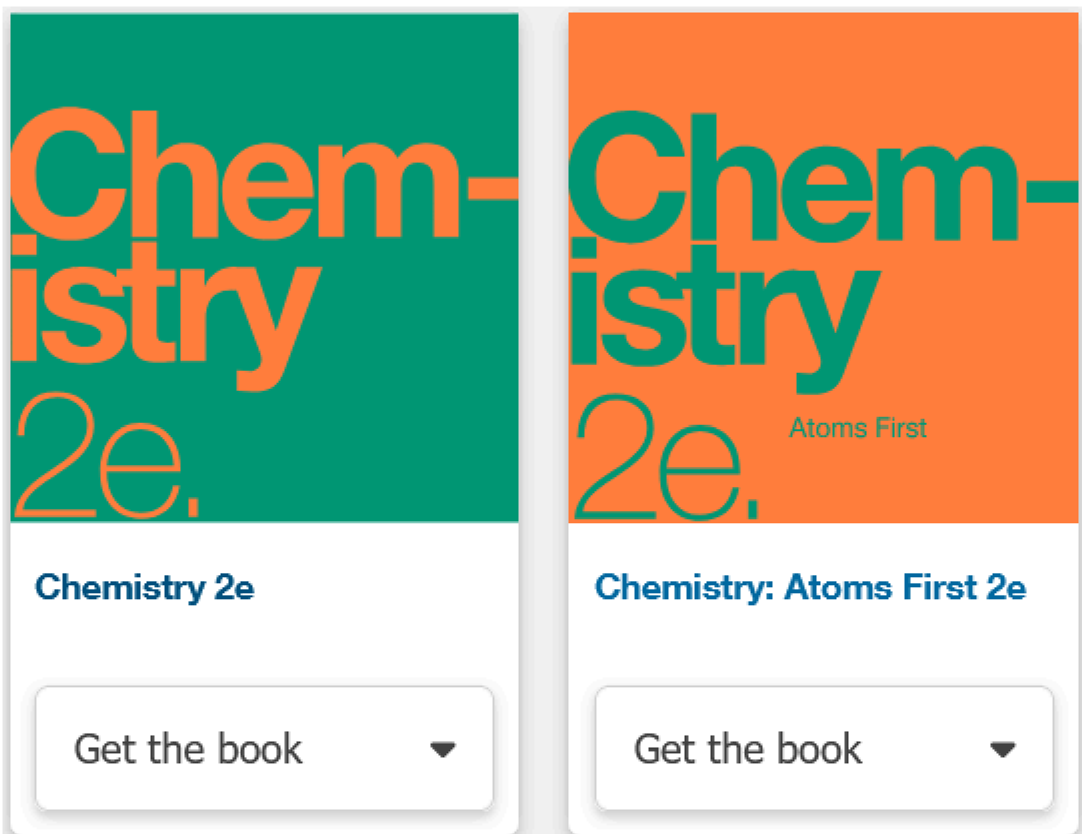
Here are some errata for the OpenStax Chemistry 2e book

https://openstax.org/details/books/chemistry-2e

Cover Image

I am partially red/green colorblind. The text of the Chemistry book cover image is a bit hard to read on the background color. I can read it, but I wonder if people with more severe color blindness would have problems. Maybe something to pass on to the accessibility team for consideration. The following image is a screenshot from the Science page: https://openstax.org/subjects/science



Ch 1

Ch 1.1

Figure 1.2 is missing alt text for part b.

Ch 1.2

Figure 1.15. I suggest adding to alt text that the water level in the hydrogen test tube is twice as far from the top as in the oxygen test tube.

Ch 1.3

Figure 1.21 - Periodic table alt text. The last sentence is incorrect and could be removed. It states “A key shows that, at room temperature, metals are solids, metalloids are liquids, and nonmetals are gases.” This misinterpretation may be due to the key being laid out in a 3 by 2 grid. In fact, the two columns of the key are unrelated, and the grid layout of the key is just an artifact of how the list of 6 unrelated key items are laid out visually. This can be seen in the image itself by noting that the color of the term Liquid in the key is blue. In the periodic table, the only two chemical symbols which are blue appear to be Hg (Mercury) and Br (Bromine). It is not true that all of the purple-background metalloid elements have blue chemical symbols, which would be the case if the last sentence of the alt text were correct. To verify my I understanding, I checked the Wikipedia article on Chemical Elements, and as of Dec 14, 2025 (https://en.wikipedia.org/w/index.php?title=Chemical_element&oldid=1324915406) it says “Most elements are solids at STP, while several are gases. Only bromine and mercury are liquid at 0 degrees Celsius (32 degrees Fahrenheit) and 1 atmosphere pressure;”

Ch 1.4

Figure 1.25 alt text - Sentence 2 says “This larger cube is made up of many smaller cubes in a 10 by 10 pattern.” Since it is a 3 dimensional cube, perhaps the end of the sentence should be “a 10 by 10 by 10 pattern.” The final sentence says “A one cubic centimeter cube is about the same width as a dime, which has a width of 1.8 centimeter.” This is correct, however the first sentence starts with “Figure A shows…” Since the last sentence describes the (b) part of the figure, perhaps the last sentence should start with “Figure B shows…”

Ch 1.5

In the third unlabelled in-text image after Figure 1.26, the figure showing significant figures in 70.607 mL and 0.00832407 mL, in the alt text for that figure, the first number is written in the alt text as “70.607 milliliters”. The second number is written in the alt text as “0.00832407 M L”. Isn’t capital “M” the abbreviation for Mega-, not milli-? Shouldn’t the second number be written in the alt text as “0.00832407 m L”, or better yet, “0.00832407 milliliters” for consistency with the first number?

Example 1.4, for the image after the phrase “Answer is 65 g (round to the ones place; no decimal places)”, in the alt text, the last sentence includes a grammatical error “...should be round to…”, which should probably be “...should be rounded to…” with an -ed suffix on “rounded”.

Ch 1.6

Figure 1.28, I would suggest two small edits to the alt text.
Suggestion 1 - The diagram shows 3 thermometers. The first sentence of the alt text says “A thermometer is shown for the Fahrenheit, Celsius and Kelvin scales.” I suggest changing the first two words of the first sentence to “Three thermometers”.
Suggestion 2 - The diagram explicitly shows 100 kelvins between the freezing and boiling points of water on the Kelvin thermometer just as it shows 100 Celsius degrees on the Celsius thermometer. The alt text omits the text describing 100 kelvins. After the alt text sentence reading “Under the kelvin scale, the boiling point of water is 373.15 K, while the freezing point of water is 273.15 K.”, I suggest inserting a new sentence like “Therefore, there are 100 kelvins between the boiling point and freezing point of water.” In addition to emphasizing the same unit sizes of the Celsius and Kelvin scales, this would emphasize the terminology difference between “100 Celsius degrees” and “100 kelvins”. Thank you.

Ch 1 key terms

Dimensional Analysis - The description doesn’t say what it is, just that its applicability includes mathematical operations from the simple to the complex. Suggest at least mentioning the idea of applying the same mathematical operations to the units as you do to the numbers.

Ch 1 Exercises

Ch1 Exercise 37. Could the answers to (f) and (g) be km2 and mL just as validly as m2 and m3? Rationale: Question says base or derived units. Chapter 1.4, Derived SI Units, Volume subsection includes mL in the text. If m3 is a derived unit but mL is technically not, shouldn’t that be mentioned in the text? Could Canadian readers also reasonably put Hectares for (f)?

Ch 1 Exercise 41. Answer to part (d) is missing from the answer key.

Ch 1 Exercise 55. I assume the point of this exercise is to ensure the student knows the conceptual difference between precision and accuracy. However, the “centers of mass” of the points for archers Y and Z are very hard to eyeball. This turns b and c, the questions dealing with accuracy, into exercises in eyeballing the center of the points rather than exercises in the conceptual understanding of accuracy. I suggest revising the images to make it more obvious which of W, Y, and Z has a center closest to the bullseye.

Ch. 2

Ch 2.1

Table 2.1 - Inconsistency. In the last row, the third column shows sample C as having 3.64 g of Hydrogen. However, in the fourth column of the same row, the denominator in the left side of the equation shows sample C as having 3.63 g of hydrogen. Note that the value 3.64 g results

in the entire equation in the fourth column being correct. The value 3.63 g results in an incorrect equation in the fourth column of the last row.

Ch 2.2

Figure 2.10 alt text - Suggestion: Sentence 5 reads “Most of the alpha particles are not deflected, but pass straight through the foil because they travel between the gold atoms.” However, the right side of the figure shows some alpha particles passing straight through which are passing within the gold atoms’ outer electron shells but between the gold atoms’ nuclei. Specifically, looking on the far right edge of the diagram and counting arrow heads down from the top, the alpha particle lines attached to the 3rd an 4th arrow heads from the top exhibit this behavior. Would it be more accurate to replace the last phrase of sentence 5, “between the gold atoms” with “between the gold atom nuclei”?

Ch 2.3

Figure 2.15 alt text - Throughout the alt text, the chemical symbol for zirconium is given as “Z R” (capital Z, space, capital R). However, in the actual chemical symbol on the diagram, the R is in lower case. Is this difference intentional to work better with assistive technologies, or is it a mistake? In other alt text I see the second letter of chemical symbols in lowercase. Thank you.

Ch 2.4

Figure 2.24 alt text - Sentence 2 reads “This molecule has a carbon atom which forms a double bond with a C H subscript 2 group and a C H subscript 3 group.” This almost makes it sound like the carbon atom has double bonds to both the CH_2 and CH_3 groups, when the diagram only shows a single bond to the CH_3 group. I suggest replacing the end of the sentence, which reads “... and a C H subscript 3 group” with “...and a single bond with a C H subscript 3 group”.

Link to Learning at end, after Figure 2.24,the link appears to be broken as of Dec-25-2025. It takes you to a general NSF news page, https://www.nsf.gov/news. The word “carvone” appears nowhere on the news page, and searching for the word “carvone” in the search box on the news page returns no results. Went to https://www.nsf.gov/news/mmg/mmg_disp.jsp?med_id=69373

Ch 2.5

First paragraph after figure 2.25, third to last sentence, should “were numerals” be “were Roman numerals”?

Figure 2.26 and Appendix A, Figure A1: In the periodic table alt text, some of the elements in the final row of the periodic table still have old names and chemical symbols, like ‘Group 13 contains “113; ununtrium; U u t; not indicated; solid; and 285.”’ The figure shows it as nihonium; Nh. Other elements toward the end of that row are similarly out of date in the alt text.

Figure 2.29 alt text. The sentence for group 11 uses capital letters for the second letters of the chemical symbols for copper. In other words, since the group 10 sentence starts “Group ten contains N i superscript 2 plus sign”, with a lowercase letter “i” as the second letter of “N i”, shouldn’t the group 11 sentence use lowercase “u” as the second letter of “C u” in the two places the chemical symbol for copper appears in the sentence? It currently starts like this, wth capital “U”’s: “Group 11 contains C U superscript plus sign and C U superscript 2 plus sign in period 4”

Ch 2.7

Figure 2.32 alt text. Sentence 2 reads “Figure B shows a 3-D ball-and-stick model of chromate.” Since figure B shows two models, should sentence 2 read, “Figure B shows 3-D ball-and-stick models of chromate and dichromate”? Sentence 3 reads “Chromate has a chromium atom at its center that forms bonds with four oxygen atoms each.” Should the final word “each” in sentence 3 be omitted? Sentence 3 deals with only one chromium to bind to oxygen atoms.

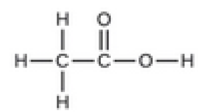
Ch2 - Exercises

Ex 2.20. Part of the instructions say to ‘select the “Mix Isotopes” tab’. However, the user interface choices look more like buttons than tabs, at least on my browser. More importantly, the button for isotope mixing is just called “Mixtures”, not “Mix Isotopes”.

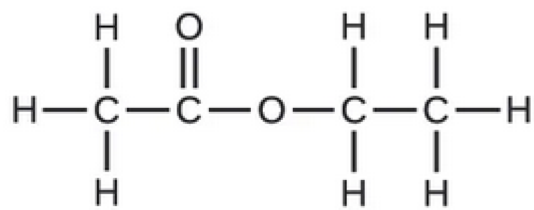
Exercises 2.29, 2.30, and 2.33(a) figures. Also the figures in the solution to exercise 2.35. All of these structural formulas figures appear much smaller than structural formulas in the rest of the book. For example, compare the small figure in Exercise 2.33(a) with the normal figure in Exercise 2.33(b), or compare all small ones listed above with the normal one in Figure 2.23 of Ch 2.4 in the main text.

33. Write the empirical formulas for the following compounds:

(a)



(b)



Ex 2.29(c) alt text. Sentence 2 reads “Each carbon atom forms a single bond with two hydrogen atoms.” Shouldn’t “a single bond” be replaced with “single bonds”? Each carbon atom has two single bonds to hydrogen, not one.

Ex 2.30(a) alt text. Sentence 4 reads “The leftmost carbon also forms single bonds with two hydrogen.” Shouldn’t sentence 4 end with “two hydrogen atoms.” for consistency with the other sentences? The final sentences read “The third carbon in the chain forms a single bond with two hydrogen atoms each. The rightmost carbon forms a single bond with three hydrogen atoms each." In the sentences about the third and rightmost carbon atoms, shouldn’t “a single bond” be replaced with “single bonds”? Also, why do the sentences about the third and rightmost carbon atoms end in the word “each”? The subject in both sentences is a single carbon atom. Also, the sentence about the second carbon atom does not end with the word “each”.

Ex 2.30(c) alt text. Sentence 1 reads “Figure C shows a structural diagram of two silicon atoms are bonded together with a single bond.” I believe the word “are” should be removed from sentence 1 for grammatical reasons. Sentence 2 reads “Each of the silicon atoms form single bonds to two chlorine atoms each and one hydrogen atom.” In sentence 2, why are the “two chlorine atoms” qualified with “each” but the “one hydrogen atom” is not qualified with “each”? Shouldn’t the “each” qualifier be on both or neither? (Neither might work because sentence 2 already starts with the word “Each” referring to the silicon atoms.) Note that other structural formulas in the Chapter 2 exercises used the word “each” in contexts I found odd, but this one seemed both add and inconsistent.

Ex 2.33(a) alt text. Sentence 2 reads “The left carbon atom forms single bonds with hydrogen atoms each.” Shouldn’t “single bonds with hydrogen atoms each.” instead read “single bonds with three hydrogen atoms.” The word “three” is missing.

Ex 2.34 and 2.35. Part of the instructions say to ‘select the “Larger Molecules” tab’. However, the user interface choices look more like buttons than tabs, at least on my browser. More importantly, the button is called “Playground”, not “Larger Molecules”.

Ex 2.53(f). Where can we find the charge of a Gallium ion to use to deduce the chemical formula? Gallium is not neaf either end of the periodic table, nor is Gallium in Figure 2.29 in Ch 2.6.

Ch 2 Exercise Solutions

Solution to ex 2.35(a), alt text. The first sentence says “A Lewis Structure is shown.” The term “Lewis structure” is not introduced until Ch 7. Would “A structural formula is shown.” be better? The final sentence ends “...and one oxygen atoms.” I believe “atom” should be singular. Also, unlike the alt text for other structural formulas, bonds are never specified as being single or double. Should that be specified?

Solution to ex 2.35(b), alt text. The first sentence says “A Lewis Structure is shown.” The term “Lewis structure” is not introduced until Ch 7. Would “A structural formula is shown.” be better? Also, unlike the alt text for other structural formulas, bonds are never specified as being single or double. Should that be specified?

Ch 3

Ch 3.1

Example 3.5, the Check Your Learning part below the main example. Could you please double check the math and especially the rounding. Taking 15.00 / 197.0 x 6.022 x 10^23 I get 4.585279 x 10^22, which rounds to 4.585 x 10^22, not 4.486 x 10^22. The end result does not change if I round between steps or not. The end result is the same if I use the fuller version of Avogadro’s number, 6.02214076 x 10^23, or the rounded version, 6.022 x 10^23.

Example 3.6. Why is the answer rounded to 3 significant digits, 0.378 mol, instead of keeping 4 significant digits, 0.3776 mol.

Ch 3.1 How Sciences Interconnect. The url www.whitehouse.gov/share/brain-initiative is not a link, and furthermore, the url is no longer valid, returning a 404 not found error when pasted directly in a browser URL bar. (I guess they should have named it the Trump’s Brain Initiative?)

Ch 3.3

Example 3.15. Why is the example answer, .004 mol sugar, rounded to 1 significant digit instead of 2 significant digits, as .0038 mol sugar? The only single digit quantity in the equation is the 1 in 1 L/1000 mL, but that is an exact, defined number, not a measured quantity.

Appendix A

Figure A1 (same issue as Figure 2.26): In the periodic table alt text, some of the elements in the final row of the periodic table still have old names and chemical symbols, like 'Group 13 contains "113; ununtrium; U u t; not indicated; solid; and 285."' The figure shows it as nihonium; Nh. Other elements toward the end of that row are similarly out of date in the alt text.

In Appendix A, would it make sense to add a link to an accessible periodic table navigable by rows and columns, in addition to having alt text on Figure A1? I believe some are linked here: https://www.perkins.org/resource/accessible-periodic-table-options/

Deans Notes to Self (OpenStax please ignore)

I did not review the alt text of Dmitri Mendeleev's original periodic table in Figure 2.25(b) of Ch 2.5 to preserve my sanity.

There may be some other odd uses of the word "each" in the alt text for additional structural formulas, at least in the Chapter 2 exercises, but that may not be worth submitting.

Continue reviewing body at beginning of Ch 3.3, Example 3.16

Continue reviewing alt text at beginning of Ch 3.